How Far Can a Small Model Go? Fine-Tuning FLAN-T5 for African Medical Question Answering

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Project links

- **GitHub repository:** https://github.com/nehemmy55/Summative-project-I
- **Demo video:** [TODO: paste video URL]
- **Live system:** [TODO: paste deployed app URL]

Abstract

Large language models answer medical exam questions well, but they are expensive to run and are weaker on African medical content. This project studies the opposite end of the scale: how well can a model small enough to train and host for free answer health questions from the African context? I fine-tune FLAN-T5-small (77M parameters) and FLAN-T5-base (248M) on AfriMed-QA, a dataset of medical questions written in Nigeria, Ghana, Kenya, Malawi and South Africa, using full fine-tuning and LoRA, with MedQuAD as auxiliary data and a TF-IDF retriever that supplies context. On 496 held-out AfriMed-QA questions, fine-tuning raises ROUGE-L on short-answer questions from 7.4 (zero-shot) to about 30, above a retrieval baseline at 22.3. On multiple-choice questions, however, no configuration is convincingly above the 20% chance level: the best run scores 24.9% (95% CI 20.4–29.2), and an earlier run of the same configuration scored 23.3%. Retrieval helps only when the knowledge base holds a close match. A separate classifier that refuses non-health questions reaches 97.4% accuracy. The main conclusion is that small models learn the *form* of medical answers but not the medical knowledge needed to choose between plausible options, so they are not safe as stand-alone medical assistants. The system is deployed as a Streamlit web application.

1. Introduction

The World Health Organization (2006) reported that the African Region carries 24% of the global burden of disease with 3% of the world's health workers. Tools that give students, community health workers and the public fast access to medical information are therefore attractive, and large language models (LLMs) now pass US medical licensing-style exams (Singhal et al., 2023). Two obstacles stand between those results and use in African settings. First, most medical benchmarks are built from US or Chinese sources, and Olatunji et al. (2025) showed that LLM accuracy drops on questions written in African medical schools. Second, the strongest models need hardware or paid APIs that many institutions cannot rely on; Olatunji et al. found that "smaller edge-friendly LLMs struggle to achieve a passing score".

This project investigates the low-resource end of that trade-off. The objective is to build and deploy a domain-specific question-answering assistant for health questions in the African context using a model that can be trained and hosted at no cost, and to measure honestly what such a model can and cannot do. The research questions are:

1. Does fine-tuning a small pretrained model improve over simple baselines on African medical questions?

2. Which choices matter: the adaptation method (LoRA or full fine-tuning), auxiliary data, retrieved context, or model size?
3. How should the system treat questions outside its domain?

English is used because it is the language of medical education and of the available data in the five countries represented; this is discussed as a limitation in Section 9. The intended users are medical and nursing students revising exam-style questions, and the system is presented as an educational prototype rather than a source of medical advice.

2. Related Work

Medical question answering. Medical QA is commonly evaluated with multiple-choice exam questions such as MedQA, drawn from US, Chinese and Taiwanese licensing exams (Jin et al., 2021), and MedMCQA from Indian entrance exams (Pal et al., 2022). MedQuAD offers 47,457 consumer-health question-answer pairs from US National Institutes of Health websites and was built for retrieval-based answering (Ben Abacha & Demner-Fushman, 2019). Recent systems rely on very large models: Med-PaLM reached a passing score on USMLE-style questions (Singhal et al., 2023).

African medical QA. AfriMed-QA is the first large pan-African medical QA dataset, with about 15,000 questions from more than 60 medical schools in 16 countries and 32 specialties (Olatunji et al., 2025). Its authors evaluated 30 LLMs and found that performance lags behind MedQA, varies by specialty and country, and that biomedical LLMs underperform general ones. The result files released with the dataset report accuracies of 53.5% for BioMistral-7B, 60.0% for Llama-3-8B and 80.4% for Llama-3-70B on 3,000 multiple-choice questions. AfriMed-QA was designed for evaluating LLMs; little work uses it to *train* small models, which is the gap this project addresses. More broadly, African NLP research has been limited by data scarcity (Nekoto et al., 2020), and question answering resources such as AfriQA cover general rather than medical knowledge (Ogundepo et al., 2023).

Models and adaptation. The Transformer (Vaswani et al., 2017) underlies T5, which casts every task as text-to-text (Raffel et al., 2020); FLAN-T5 further fine-tunes T5 on over a thousand instruction-phrased tasks (Chung et al., 2022). LoRA freezes the pretrained weights and trains low-rank update matrices, cutting trainable parameters by orders of magnitude (Hu et al., 2022). Roberts et al. (2020) showed that the amount of factual knowledge a T5 model can recall without external context grows with its size, which suggests small models will be limited on knowledge-heavy questions. Retrieval-augmented generation addresses this by giving the model retrieved text to condition on (Lewis et al., 2020).

3. Dataset and Data Preparation

3.1 Sources

All data is naturally occurring, human-written text; no synthetic or LLM-generated data was used.

Dataset	Role	Portion used	Licence
AfriMed-QA v2.5 (Olatunji et al., 2025)	Main dataset	3,723 single-answer multiple-choice questions (MCQ) and 1,236 short-answer questions (SAQ)	CC BY-NC-SA 4.0
AfriMed-QA consumer queries	Domain gate only	10,000 health questions from the public	CC BY-NC-SA 4.0
MedQuAD (Ben Abacha & Demner-Fushman, 2019)	Auxiliary training data; retrieval passages	3,000 train and 200 test pairs, plus passages, from 7 NIH sources	CC BY 4.0
WebQuestions (Berant et al., 2013); TREC (Li & Roth, 2002)	Domain gate, non-health class	11,673 general questions	Research use

AfriMed-QA was downloaded from the authors' public GitHub repository. Of its 15,275 rows, the 10,000 consumer queries have no reference answer, which I only discovered when inspecting the file; they cannot be used to train or evaluate a QA model, so I used them instead as realistic examples of health questions for the domain gate. The answered portion comes from Nigeria (2,176), Ghana (1,491), Kenya (776), Malawi (461) and South Africa (54); the largest specialties are general surgery, obstetrics and gynaecology, and paediatrics. MCQs have five options on average; the correct option averages 4 words, while SAQ reference answers average 43 words and run up to 900.

MedQuAD was added because AfriMed-QA has fewer than 5,000 answered questions. The two rare-genetic-disease sources (GHR, GARD) were excluded as irrelevant to primary care. Whether this US data helps is tested as an experiment (Section 6).

3.2 Preprocessing

1. **Text cleaning:** whitespace and line breaks collapsed; characters corrupted by a bad encoding removed; exam numbering ("31.") stripped from questions; enumeration labels ("a.", "B)") stripped from options only when the letter matches the option's position, so that answers such as "E. coli" are preserved.
2. **Filtering:** 315 MCQs with more than one correct option and one duplicate question were removed.
3. **Answer shortening:** reference answers were cut at sentence boundaries to about 45 words, since a small model cannot generate multi-paragraph explanations.
4. **Format:** each example becomes a text prompt. For MCQs the options are listed as "(A) … (B) …" and the target is the *text* of the correct option rather than its letter.

3.3 Split

AfriMed-QA's own split is designed for benchmarking (97% of MCQs are in its test portion, and its training portion is crowdsourced while its test portion is expert-written). I therefore pooled the answered rows and made a stratified 80/10/10 split by question type with a fixed seed.

Split	AfriMed MCQ	AfriMed SAQ	MedQuAD	Total
Train	2,978	989	3,000	6,967
Validation	372	124	–	496
Test	373	123	200	696

A retrieval knowledge base of 8,221 passages was built from training answers and unused MedQuAD passages; validation and test answers are excluded.

3.4 Data quality and limitations

The data shows the difficulties of real text: missing spaces and misspellings from digitised exam papers ("thefollowing", "Ceplhalic vein"), placeholder options ("n/a"), and some answer keys that appear incorrect. Many SAQs follow templates ("What is the primary method used for diagnosing X in Africa?"), so near-duplicate questions occur across the random split: 4.9% of test SAQs, 2.4% of test MCQs and 14.5% of MedQuAD test questions have a training question with TF-IDF cosine similarity of at least 0.8. No exact question is shared between splits. Five countries dominate, and everything is in English.

4. Methodology

The system has three stages: a domain gate, a retriever, and a generator.

Domain gate. A logistic-regression classifier over character 2–5-gram TF-IDF features decides whether a question is about health. It was trained on 11,894 health questions (AfriMed-QA) and 9,154 general questions. Character n-grams were chosen for robustness to the spelling errors noted above. The baseline for this component is a threshold on the similarity of the best retrieved passage.

Retriever. Questions and knowledge-base entries are represented with word 1–2-gram TF-IDF vectors and ranked by cosine similarity. This sparse retriever needs no training and no GPU.

Generator. FLAN-T5 is an encoder–decoder Transformer: the encoder turns the tokenised prompt into contextual embeddings with self-attention, and the decoder generates the answer token by token, attending to the encoder output. FLAN-T5-small has 77M parameters (8 layers per stack, hidden size 512) and FLAN-T5-base 248M (12 layers, hidden size 768). The model was chosen because its instruction pre-training gives a usable zero-shot baseline and because the small version trains on a laptop CPU. Training minimises token-level cross-entropy with AdamW, weight decay 0.01, 6% linear warm-up then linear decay, batch size 8, gradient clipping at 1.0, for 3 epochs; inputs are limited to 160 tokens (256 with context) and targets to 48. The checkpoint with the lowest validation loss is kept. Answers are decoded with beam search (4 beams).

Two adaptation methods are compared. *Full fine-tuning* updates all weights with learning rate 3e-4. *LoRA* freezes them and learns, for each attention query and value projection W, a low-rank update BA of rank r scaled by α/r ($\alpha = 2r$, dropout 0.05), with learning rate 1e-3; this trains 0.45% of the parameters at $r = 8$.

In the retrieval-augmented configuration the best passage is prepended to the prompt as "Context: …". During training a question is prevented from retrieving its own entry, otherwise the model would simply learn to copy.

Baselines. (1) *TF-IDF retrieval*: return the stored answer of the most similar knowledge-base question, a classical FAQ-style system with no learning. (2) *Zero-shot FLAN-T5-small*: the pretrained model without fine-tuning. A random-option baseline gives the chance level for MCQs.

Metrics. MCQ *accuracy* is computed by mapping the generated text to the closest option (exact match, then option letter, then highest token overlap); chance is 20%. Open answers are scored with *token-level F1* (Rajpurkar et al., 2016) and *ROUGE-L* (Lin, 2004), which give partial credit for overlapping content; exact match is reported in the repository but is almost always zero for free text. The domain gate is scored with accuracy, precision, recall and F1. 95% confidence intervals use 2,000 bootstrap resamples of the test set.

Implementation. The code uses PyTorch, Hugging Face Transformers (Wolf et al., 2020) and PEFT, and scikit-learn (Pedregosa et al., 2011). I implemented the data pipeline, training loop, retrieval-augmented setup, metrics, domain gate, analysis and web app; datasets, pretrained checkpoints and libraries are reused.

5. Experimental Setup

Each experiment changes one factor relative to an earlier one. E1–E2 were trained on a 4-core laptop CPU (roughly 1.5–2 hours each); E3–E7 used the same code on a Colab T4 GPU (4–18 minutes each).

ID	Change	Question
E1	LoRA r = 8, AfriMed-QA	Does fine-tuning beat the baselines?
E2	LoRA r = 32	Is adapter capacity the bottleneck?
E3	Full fine-tuning	Is LoRA enough, or do all weights need updating?
E4	E3 + MedQuAD training data	Does auxiliary consumer-health data help?
E5	E4 + retrieved context	Does supplying knowledge at inference help?
E6	E3 with FLAN-T5-base	Does a 3× larger model help?
E7	E5 with FLAN-T5-base	Best combination

6. Experiments and Results

Table 1. Test-set results. MCQ accuracy with 95% bootstrap CI; F1 and ROUGE-L in %.

System	Trainable params	MCQ accuracy	SAQ F1	SAQ ROUGE-L	MedQuAD F1	MedQuAD ROUGE-L
Random option	–	16.9 [13.1, 20.9]	0.0	0.0	0.0	0.0
Baseline: TF-IDF retrieval	–	8.6 [5.6, 11.5]	25.4	22.3	19.7	17.0
Baseline: zero-shot FLAN-T5-small	–	19.3 [15.3, 23.6]	7.5	7.4	9.5	9.2
E1 Small, LoRA r=8	344,064	23.6 [19.3, 28.1]	32.0	27.8	19.5	18.9
E2 Small, LoRA r=32	1,376,256	23.1 [18.8, 27.4]	31.9	27.9	19.5	18.6
E3 Small, full fine-tuning	76,961,152	22.8 [18.2, 27.1]	33.4	29.9	18.1	17.2
E4 Small, full, + MedQuAD	76,961,152	22.0 [17.7, 26.3]	32.8	28.8	26.7	24.5
E5 Small, full, + MedQuAD + retrieval	76,961,152	24.9 [20.4, 29.2]	33.0	30.1	31.4	29.3
E6 Base, full fine-tuning	247,577,856	20.4 [16.4, 24.4]	35.4	31.6	18.1	18.3
E7 Base, full, + MedQuAD + retrieval	247,577,856	22.5 [18.2, 26.8]	34.2	31.2	34.1	31.6

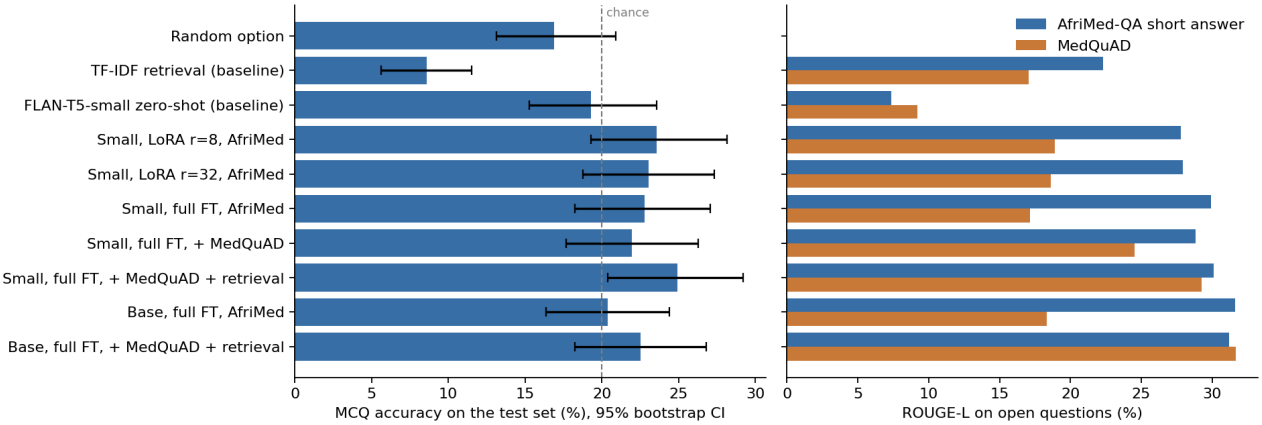


Figure 1. MCQ accuracy with 95% confidence intervals (left) and ROUGE-L on open questions (right).

Fine-tuning versus baselines (RQ1). On open questions fine-tuning is clearly effective: SAQ ROUGE-L rises from 7.4 zero-shot to 27.8 with LoRA, passing the retrieval baseline (22.3). The zero-shot model mostly echoes the topic of the question. On MCQs the picture is different. Zero-shot accuracy is 19.3%, and the fine-tuned models fall between 20.4% and 24.9%; six of the seven confidence intervals contain 20%. A one-sided binomial test against chance gives $p = 0.012$ for the best model (E5, 93/373) and $p \geq 0.05$ for all others. With seven models tested, $p = 0.012$ does not survive a Bonferroni correction (threshold 0.007), and E5 was trained twice with identical settings: the first run scored 23.3% (87/373, $p = 0.064$) and the second, which is the one reported and deployed, 24.9%. The 1.6-point gap between two identical runs is as large as most differences between configurations, so I do not

treat any model as reliably better than chance. The retrieval baseline scores *below* chance (8.6%) only because 70.5% of its retrieved answers match no option and are counted as wrong.

LoRA versus full fine-tuning (E1–E3). Quadrupling the LoRA rank changes nothing (27.8 vs 27.9 ROUGE-L), so adapter capacity is not the limit. Full fine-tuning lowers validation loss from 1.78 to 1.58 and adds about 2 ROUGE-L points while training 220 times more parameters. LoRA therefore recovers most of the benefit with a 1.4 MB adapter, consistent with Hu et al. (2022).

Auxiliary data (E4). Adding MedQuAD raises MedQuAD ROUGE-L from 17.2 to 24.5 but leaves AfriMed SAQ unchanged (29.9 → 28.8) and MCQ accuracy unchanged. US consumer-health text teaches the model that style of answer; it does not transfer to African clinical exam questions.

Retrieved context (E5). Retrieval raises MedQuAD ROUGE-L to 29.3 and SAQ ROUGE-L slightly to 30.1. Figure 2 shows why the effect is uneven: the gain appears only when the retrieved entry is similar to the question (TF-IDF similarity above 0.4) and is absent or negative below 0.2. Most AfriMed test questions (94 of 123) have a best match below 0.4.

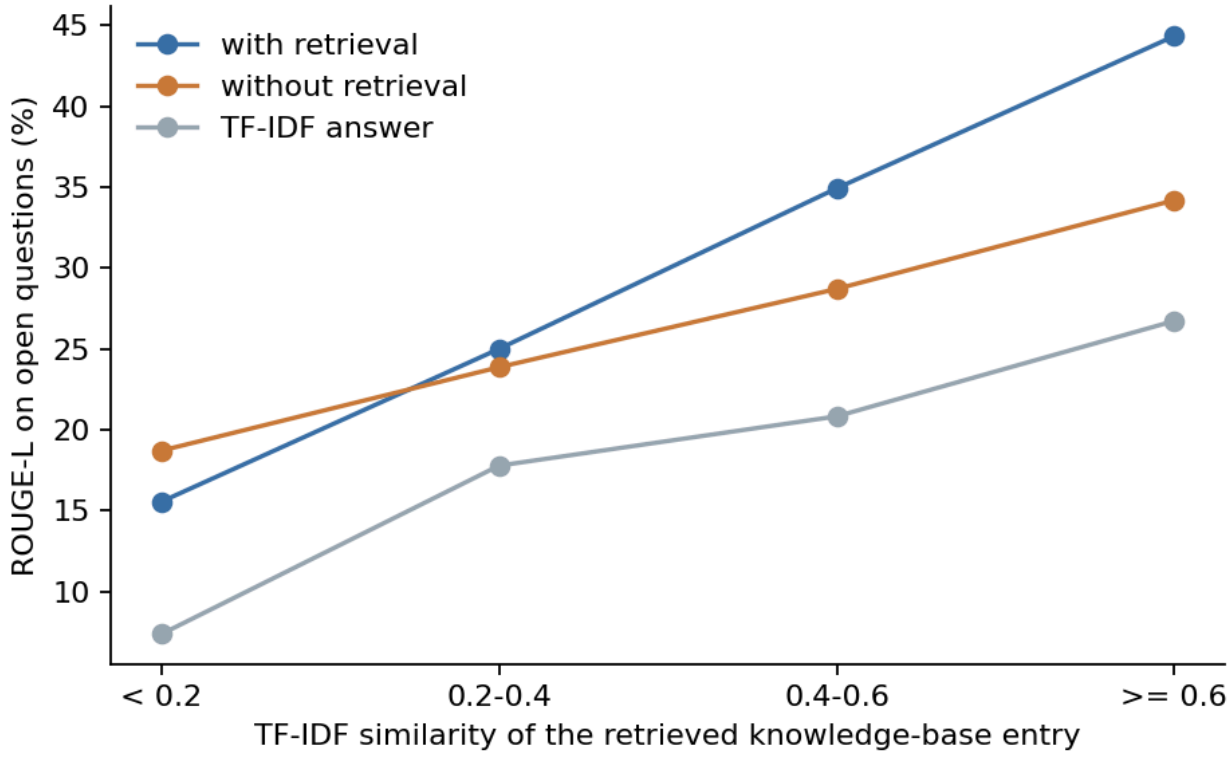


Figure 2. ROUGE-L on open questions by similarity of the retrieved entry, with and without retrieved context.

Model size (E6–E7). FLAN-T5-base reaches a much lower validation loss (1.34 against 1.58) and slightly better open-answer scores (+1.7 ROUGE-L on SAQ), yet its MCQ accuracy is the lowest of the fine-tuned models (20.4%). Lower loss here reflects more fluent reproduction of option text, not better selection of the correct option. A 3× increase in parameters is far too little to close the gap to the 7–70-billion-parameter models that Olatunji et al. (2025) report at 54–80%.

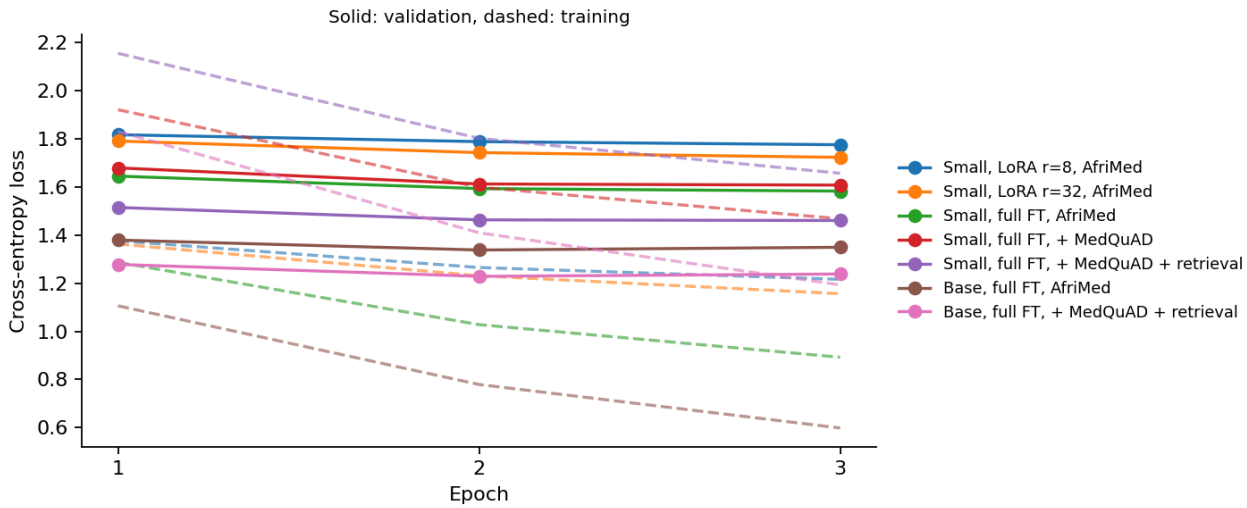


Figure 3. Training and validation loss. Validation loss is averaged per token and training loss per batch, so the two are not directly comparable in level; the widening gap for the AfriMed-only full fine-tuning runs indicates overfitting.

Domain gate (RQ3). The classifier reaches 97.4% accuracy (precision 99.8%, recall 94.7%, F1 97.2%) on 4,748 held-out questions, against 68.6% for the similarity threshold.

Final model. E5 is deployed. It is within noise of the larger E7 on AfriMed questions, is three times smaller, and runs on free CPU hosting.

7. Error Analysis and Discussion

7.1 Multiple-choice questions

Where the deployed model is right, the question usually has a short, distinctive answer, for example choosing “Galactocele” for a breastfeeding woman with a lump that yields white fluid, or “Cephalic vein” as the guide to the deltopectoral groove.

Where it is wrong, three patterns stand out (Figure 4):

- **Position bias.** The model picks option C for 27.9% of questions although C is correct for 20.4%, and picks E for 12.1% against 15.5%. Selection bias of this kind is documented even in large LLMs (Zheng et al., 2024). Part of it is learned from the data: in the training MCQs, C is the most frequent correct position (23.8%) and E the least (11.0%).
- **Long options.** Accuracy is 28.6% when the correct option has 1–2 words but 17.2% when it has six or more, such as full statements in “which of the following is true” questions.
- **Shared mistakes.** 43.2% of test questions are answered correctly by none of the seven fine-tuned models, where independent guessing would give about 21%. The models are not guessing randomly; they follow the same surface cues, which are right about as often as chance.

Negated questions (“all of the following except”) are not harder for the model (28.3% against 23.5% for the rest, a difference within noise for 113 questions), which suggests it is not reading the question closely enough for negation to matter.

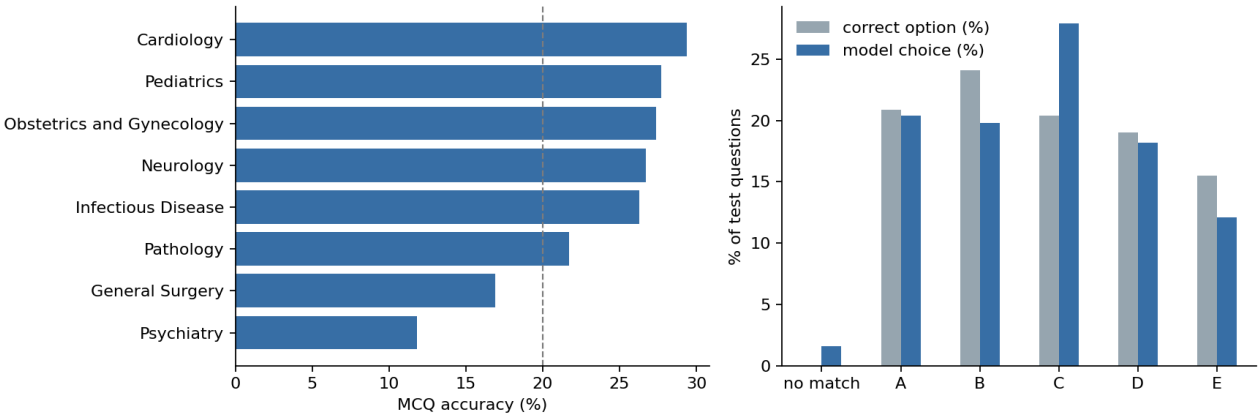


Figure 4. MCQ accuracy by specialty and distribution of the selected option position for the deployed model.

A typical failure: for a 19-year-old with haematuria, hearing loss and an affected brother, the model answers "autosomal dominant polycystic kidney disease" instead of Alport syndrome. The answer is a plausible renal diagnosis but ignores the decisive clue, which requires knowledge the model does not have.

7.2 Open questions

Of the 123 AfriMed short answers from the deployed model, an automatic categorisation gives 28 good (ROUGE-L ≥ 0.5), 26 partially correct, 28 that restate the question, and 41 wrong or unrelated.

- **Fluent hallucination.** Asked for the leading cause of death in under-fives in sub-Saharan Africa, the model writes "Cataract, a clouding of the eye's lens, is the leading cause of death among children under 5 years old in sub-Saharan Africa" (reference: malaria). The sentence is grammatical and confident, and it still earns 0.45 ROUGE-L because it repeats the question, which exposes the weakness of overlap metrics for medical correctness.
- **Restating the question.** Instructions such as "Discuss …" or "Write short notes on …" are often echoed back ("Surgical management of inguinal hernia"). These need long structured answers, beyond the 48-token output limit.
- **Copying boilerplate.** The highest-scoring MedQuAD answers reproduce standard NIH sentences ("Anything that increases your risk of getting a disease is called a risk factor") found in the retrieved passage. Part of the retrieval gain on MedQuAD is therefore template matching.
- **Tokenizer gap.** The T5 vocabulary has no "<" symbol, so "Temperature < 36" is generated as "Temperature 36", making some numeric options impossible to reproduce.

7.3 Domain gate

Of 124 errors, 119 are health questions wrongly refused and 5 are general questions accepted, so the gate errs on the side of refusing. On twenty hand-written probes it accepted "hello" and "how are you today?" as health questions (no greetings exist in the training data) and was uncertain on "what is hypertension?" (0.53), because short "What is X?" questions are common in TREC. The app therefore asks for a complete question when fewer than three words are entered.

7.4 Discussion

The results fit the literature. Roberts et al. (2020) found closed-book factual recall depends on model size, and Olatunji et al. (2025) found small LLMs fail AfriMed-QA; this project shows that task-specific fine-tuning of a much smaller model does not escape that limit. Fine-tuning teaches format and domain

vocabulary quickly, which is why open-answer overlap scores rise sharply, but selecting the right option among five plausible ones is a test of knowledge. Retrieval is the sensible remedy (Lewis et al., 2020), but only as good as its knowledge base: mine contains answers to *other* questions rather than a medical reference, and it rarely holds what a new exam question needs.

8. Deployment

The system is deployed as a Streamlit application. On start-up the app loads the domain-gate classifier, fits the TF-IDF retriever on the knowledge base, and loads the fine-tuned FLAN-T5-small weights from the Hugging Face Hub; these are cached so each request only runs inference. A user can (1) type a health question and receive a generated answer, or (2) paste an exam question with its options and see which option the model selects. Each answer is accompanied by the gate's probability, the retrieved knowledge-base entry with its similarity score, and a warning when similarity is low. Non-health questions receive a refusal. A banner states that the tool is an educational prototype and not medical advice. The link is given at the top of this report.

9. Limitations and Future Work

- **Reliability.** MCQ accuracy is at chance and open answers are frequently wrong; the system must not be used for clinical decisions.
- **Language.** All data is English. Kinyarwanda, Swahili, Hausa and other languages spoken by patients are not covered, and no real medical QA data exists for them yet.
- **Evaluation.** The test set has 373 MCQs, so differences of 2–3 points are within noise; configurations were run with one seed, except E5, whose two runs differed by 1.6 points of MCQ accuracy; near-duplicate templated SAQs across the split inflate open-answer scores; ROUGE and F1 measure overlap rather than correctness, and no clinician reviewed the outputs.
- **Data.** Fewer than 5,000 answered questions from five countries; some answer keys are questionable.
- **Experiments.** LoRA on FLAN-T5-base failed in Colab because of a library conflict and is not reported.

With more time and compute I would (1) replace the question-answer knowledge base with authoritative references such as WHO and national treatment guidelines and use a dense retriever; (2) fine-tune a 1–8B parameter model with LoRA and 4-bit quantisation, which the literature suggests is the minimum for non-trivial MCQ accuracy; (3) add clinician evaluation of correctness and harm; and (4) collect questions in African languages.

10. Conclusion

A FLAN-T5-small model fine-tuned on AfriMed-QA learns to produce medical-sounding short answers that overlap with expert references far more than the pretrained model's, and a simple classifier reliably keeps the assistant within its domain. But across seven configurations varying the adaptation method, data, retrieval and model size, none answered African medical exam questions convincingly better than chance. The most useful lessons were methodological: a validation loss that falls steadily can coexist with no gain on the metric that matters; a strong-looking gain from retrieval partly came from shared boilerplate; and a fluent wrong answer can score well on ROUGE. For this domain, model size and the quality of the retrieved knowledge matter more than the fine-tuning recipe.

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